

Manish Kumar

+91-9675800189 | manishsharma9675800189@gmail.com | [Linkedin](#) | [GitHub](#)

Professional Summary

Software Engineer with 2 years of experience in Embedded Linux application development using C, LVGL, Linux, and Python. Experienced in GUI development, debugging, performance optimization, CAN communication, and MISRA C compliance. Currently transitioning into AI/ML Engineering with hands-on experience in Machine Learning, NLP, Flask, LangChain, Vector Databases, and Retrieval-Augmented Generation (RAG) systems. Passionate about building intelligent applications powered by Machine Learning and Large Language Models.

Technical Skills

- **Programming Languages:** C, Python, SQL
- **Embedded Systems:** LVGL, Linux, Embedded Application Development, CAN Communication, Testing and Debugging, Performance Optimization, MISRA C Standards
- **Data Analysis:** Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering
- **Machine Learning:** Scikit-learn, Regression Models, Classification Models, Ensemble Learning (Bagging, Boosting), ML Pipeline Development, Model Deployment
- **Deep Learning:** PyTorch, ANN, CNN, RNN, LSTM, Transformer Fundamentals
- **Generative AI:** LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, LLM Integration
- **Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Scikit-learn, Flask, Pytorch
- **Web Technologies:** HTML, CSS, JavaScript
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Docker

Experience

Software Engineer

2024 – Present

Avench System Pvt. Ltd.

Bangalore, Karnataka

- Developed Embedded Linux-based GUI applications using C and LVGL graphics library for embedded devices.
- Designed and implemented responsive graphical user interfaces including widgets, icons, rollers, text components, menus, and screen alignment features.
- Integrated CAN communication for real-time data transfer and display between embedded systems and GUI applications.
- Implemented multilingual support including English, French, and German languages with dynamic language switching through application menu settings.
- Followed MISRA C coding standards to ensure code quality, maintainability, reliability, and compliance in embedded software development.
- Performed debugging, issue analysis, feature enhancement, and performance optimization for Linux-based embedded applications.
- Developed Python automation scripts to support development workflows, testing, and internal project activities.
- Collaborated with cross-functional teams to develop scalable and user-friendly embedded applications for production environments.

Projects

Email Spam Detection System — Python, Flask, Scikit-learn, NLP

2026

- Developed an **end-to-end NLP-based Email Spam Detection System** using Machine Learning classification techniques.
- Performed **text preprocessing, tokenization, stemming, stopword removal, and TF-IDF vectorization** for feature extraction.
- Built an **end-to-end Machine Learning pipeline** for data preprocessing, feature transformation, model training, and prediction workflow automation.
- Trained and evaluated **multiple classification models** and selected the best-performing model based on **F1-score evaluation metrics**.
- Integrated and deployed the trained model into a **Flask web application** using **Render** for real-time spam prediction.
- GitHub: [Email Spam Detection System](#)
- Live Demo: [Email Spam Detection System Demo](#)

- Developed an AI-powered PDF Question Answering Chatbot using Retrieval-Augmented Generation (RAG) architecture.
- Implemented PDF ingestion, text extraction, chunking, and embedding generation for semantic document understanding.
- Integrated ChromaDB vector database for embedding storage and similarity-based document retrieval.
- Built a LangChain-powered retrieval pipeline to provide context-aware answers from uploaded documents.
- Integrated Google Gemini LLM to generate responses grounded in retrieved document context and reduce hallucinations.
- Developed a Flask-based web application supporting PDF upload and interactive document-based conversations.
- GitHub: [PDF Question Answering Chatbot](#)

Education

Bachelor of Computer Applications (BCA) <i>Uttaranchal University</i>	2024 – 2027 <i>Dehradun, Uttarakhand</i>
Diploma in Information Technology and Data Science — CGPA: 9.3/10 <i>Nettur Technical Training Foundation (NTTF)</i>	2021 – 2024 <i>Bangalore, Karnataka</i>